One-time password

Registration: The service provide users with unique secret keys (using a QR code usually) to store in their mobile application

Authentication: The device generates a one-time password based on the secret stored in the device

There are two standards both based on MAC

- HOTP (HMAC-based one-time password) based on a counter
- **TOTP** (Time-based one-time password)
  based on time quantum ~ usually 30 seconds

An example of a USB Security Key : *YubiKey*

Hardware device (USB and/or Bluetooth) to provide authentication

- Support one-time password (HOTP or TOTP)
- Support digital signature (OpenPGP and PKCS#11)
- Compatible with *WebAuthn*
- Can combine a biometrics factor

